

NB, line 8). In contrast, the multinomial model estimates $\hat{P}(t|c)$ as the *fraction of tokens* or *fraction of positions* in documents of class c that contain term t (Equation (13.7)). When classifying a test document, the Bernoulli model uses binary occurrence information, ignoring the number of occurrences, whereas the multinomial model keeps track of multiple occurrences. As a result, the Bernoulli model typically makes many mistakes when classifying long documents. For example, it may assign an entire book to the class *China* because of a single occurrence of the term *China*.

The models also differ in how nonoccurring terms are used in classification. They do not affect the classification decision in the multinomial model; but in the Bernoulli model the probability of nonoccurrence is factored in when computing $P(c|d)$ (Figure 13.3, APPLYBERNOULLINB, Line 7). This is because only the Bernoulli NB model models absence of terms explicitly.



Example 13.2: Applying the Bernoulli model to the example in Table 13.1, we have the same estimates for the priors as before: $\hat{P}(c) = 3/4$, $\hat{P}(\bar{c}) = 1/4$. The conditional probabilities are:

$$\begin{aligned}\hat{P}(\text{Chinese}|c) &= (3+1)/(3+2) = 4/5 \\ \hat{P}(\text{Japan}|c) = \hat{P}(\text{Tokyo}|c) &= (0+1)/(3+2) = 1/5 \\ \hat{P}(\text{Beijing}|c) = \hat{P}(\text{Macao}|c) = \hat{P}(\text{Shanghai}|c) &= (1+1)/(3+2) = 2/5 \\ \hat{P}(\text{Chinese}|\bar{c}) &= (1+1)/(1+2) = 2/3 \\ \hat{P}(\text{Japan}|\bar{c}) = \hat{P}(\text{Tokyo}|\bar{c}) &= (1+1)/(1+2) = 2/3 \\ \hat{P}(\text{Beijing}|\bar{c}) = \hat{P}(\text{Macao}|\bar{c}) = \hat{P}(\text{Shanghai}|\bar{c}) &= (0+1)/(1+2) = 1/3\end{aligned}$$

The denominators are $(3+2)$ and $(1+2)$ because there are three documents in c and one document in $\bar{c}$ and because the constant B in Equation (13.7) is 2 – there are two cases to consider for each term, occurrence and nonoccurrence.

The scores of the test document for the two classes are

$$\begin{aligned}\hat{P}(c|d_5) &\propto \hat{P}(c) \cdot \hat{P}(\text{Chinese}|c) \cdot \hat{P}(\text{Japan}|c) \cdot \hat{P}(\text{Tokyo}|c) \\ &\quad \cdot (1 - \hat{P}(\text{Beijing}|c)) \cdot (1 - \hat{P}(\text{Shanghai}|c)) \cdot (1 - \hat{P}(\text{Macao}|c)) \\ &= 3/4 \cdot 4/5 \cdot 1/5 \cdot 1/5 \cdot (1-2/5) \cdot (1-2/5) \cdot (1-2/5) \\ &\approx 0.005\end{aligned}$$

and, analogously,

$$\begin{aligned}\hat{P}(\bar{c}|d_5) &\propto 1/4 \cdot 2/3 \cdot 2/3 \cdot 2/3 \cdot (1-1/3) \cdot (1-1/3) \cdot (1-1/3) \\ &\approx 0.022\end{aligned}$$

Thus, the classifier assigns the test document to $\bar{c} = \text{not-China}$. When looking only at binary occurrence and not at term frequency, Japan and Tokyo are indicators for $\bar{c}$ ($2/3 > 1/5$) and the conditional probabilities of Chinese for c and $\bar{c}$ are not different enough ($4/5$ vs. $2/3$) to affect the classification decision.

13.4 Properties of Naive Bayes

To gain a better understanding of the two models and the assumptions they make, let us go back and examine how we derived their classification rules in Chapters 11 and 12. We decide class membership of a document by assigning it to the class with the maximum a posteriori probability (cf. Section 11.3.2, page 226), which we compute as follows:

$$(13.9) \quad \begin{aligned} c_{\text{map}} &= \arg \max_{c \in \mathcal{C}} P(c|d) \\ &= \arg \max_{c \in \mathcal{C}} \frac{P(d|c)P(c)}{P(d)} \end{aligned}$$

$$(13.10) \quad = \arg \max_{c \in \mathcal{C}} P(d|c)P(c),$$

where Bayes' rule (Equation (11.4), page 220) is applied in (13.9) and we drop the denominator in the last step because $P(d)$ is the same for all classes and does not affect the argmax.

We can interpret Equation (13.10) as a description of the generative process we assume in Bayesian text classification. To generate a document, we first choose class c with probability $P(c)$ (top nodes in Figures 13.4 and 13.5). The two models differ in the formalization of the second step, the generation of the document given the class, corresponding to the conditional distribution $P(d|c)$:

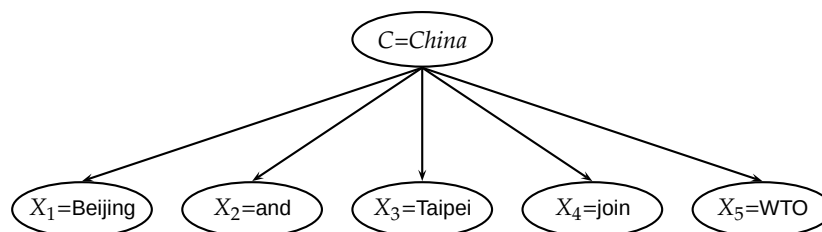
$$(13.11) \quad \textbf{Multinomial} \quad P(d|c) = P(\langle t_1, \dots, t_k, \dots, t_{n_d} \rangle | c)$$

$$(13.12) \quad \textbf{Bernoulli} \quad P(d|c) = P(\langle e_1, \dots, e_i, \dots, e_M \rangle | c),$$

where $\langle t_1, \dots, t_{n_d} \rangle$ is the sequence of terms as it occurs in d (minus terms that were excluded from the vocabulary) and $\langle e_1, \dots, e_i, \dots, e_M \rangle$ is a binary vector of dimensionality M that indicates for each term whether it occurs in d or not.

It should now be clearer why we introduced the document space $\mathbb{X}$ in Equation (13.1) when we defined the classification problem. A critical step in solving a text classification problem is to choose the document representation. $\langle t_1, \dots, t_{n_d} \rangle$ and $\langle e_1, \dots, e_M \rangle$ are two different document representations. In the first case, $\mathbb{X}$ is the set of all term sequences (or, more precisely, sequences of term tokens). In the second case, $\mathbb{X}$ is $\{0, 1\}^M$.

We cannot use Equations (13.11) and (13.12) for text classification directly. For the Bernoulli model, we would have to estimate $2^M |\mathcal{C}|$ different parameters, one for each possible combination of M values e_i and a class. The number of parameters in the multinomial case has the same order of magni-



► **Figure 13.4** The multinomial NB model.

tude.³ This being a very large quantity, estimating these parameters reliably is infeasible.

CONDITIONAL
INDEPENDENCE
ASSUMPTION

To reduce the number of parameters, we make the Naive Bayes *conditional independence assumption*. We assume that attribute values are independent of each other given the class:

$$(13.13) \quad \textbf{Multinomial} \quad P(d|c) = P(\langle t_1, \dots, t_{n_d} \rangle | c) = \prod_{1 \leq k \leq n_d} P(X_k = t_k | c)$$

$$(13.14) \quad \textbf{Bernoulli} \quad P(d|c) = P(\langle e_1, \dots, e_M \rangle | c) = \prod_{1 \leq i \leq M} P(U_i = e_i | c).$$

RANDOM VARIABLE X

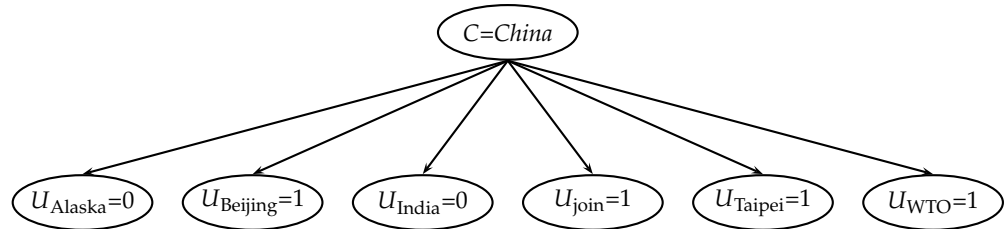
RANDOM VARIABLE U

We have introduced two random variables here to make the two different generative models explicit. X_k is the random variable for position k in the document and takes as values terms from the vocabulary. $P(X_k = t | c)$ is the probability that in a document of class c the term t will occur in position k . U_i is the random variable for vocabulary term i and takes as values 0 (absence) and 1 (presence). $P(U_i = 1 | c)$ is the probability that in a document of class c the term t_i will occur – in any position and possibly multiple times.

We illustrate the conditional independence assumption in Figures 13.4 and 13.5. The class *China* generates values for each of the five term attributes (multinomial) or six binary attributes (Bernoulli) with a certain probability, independent of the values of the other attributes. The fact that a document in the class *China* contains the term Taipei does not make it more likely or less likely that it also contains Beijing.

In reality, the conditional independence assumption does not hold for text data. Terms *are* conditionally dependent on each other. But as we will discuss shortly, NB models perform well despite the conditional independence assumption.

3. In fact, if the length of documents is not bounded, the number of parameters in the multinomial case is infinite.



► **Figure 13.5** The Bernoulli NB model.

Even when assuming conditional independence, we still have too many parameters for the multinomial model if we assume a different probability distribution for each position k in the document. The position of a term in a document by itself does not carry information about the class. Although there is a difference between *China sues France* and *France sues China*, the occurrence of China in position 1 versus position 3 of the document is not useful in NB classification because we look at each term separately. The conditional independence assumption commits us to this way of processing the evidence.

Also, if we assumed different term distributions for each position k , we would have to estimate a different set of parameters for each k . The probability of bean appearing as the first term of a *coffee* document could be different from it appearing as the second term, and so on. This again causes problems in estimation owing to data sparseness.

POSITIONAL
INDEPENDENCE

For these reasons, we make a second independence assumption for the multinomial model, *positional independence*: The conditional probabilities for a term are the same independent of position in the document.

$$P(X_{k_1} = t|c) = P(X_{k_2} = t|c)$$

for all positions k_1, k_2 , terms t and classes c . Thus, we have a single distribution of terms that is valid for all positions k_i and we can use X as its symbol.⁴ Positional independence is equivalent to adopting the bag of words model, which we introduced in the context of ad hoc retrieval in Chapter 6 (page 117).

With conditional and positional independence assumptions, we only need to estimate $\Theta(M|C|)$ parameters $P(t_k|c)$ (multinomial model) or $P(e_i|c)$ (Bernoulli

4. Our terminology is nonstandard. The random variable X is a categorical variable, not a multinomial variable, and the corresponding NB model should perhaps be called a *sequence model*. We have chosen to present this sequence model and the multinomial model in Section 13.4.1 as the same model because they are computationally identical.

► **Table 13.3** Multinomial versus Bernoulli model.

	multinomial model	Bernoulli model
event model	generation of token	generation of document
random variable(s)	$X = t$ iff t occurs at given pos	$U_t = 1$ iff t occurs in doc
document representation	$d = \langle t_1, \dots, t_k, \dots, t_{n_d} \rangle, t_k \in V$	$d = \langle e_1, \dots, e_i, \dots, e_M \rangle,$ $e_i \in \{0, 1\}$
parameter estimation	$\hat{P}(X = t c)$	$\hat{P}(U_i = e c)$
decision rule: maximize	$\hat{P}(c) \prod_{1 \leq k \leq n_d} \hat{P}(X = t_k c)$	$\hat{P}(c) \prod_{i \in V} \hat{P}(U_i = e_i c)$
multiple occurrences	taken into account	ignored
length of docs	can handle longer docs	works best for short docs
# features	can handle more	works best with fewer
estimate for term the	$\hat{P}(X = \text{the} c) \approx 0.05$	$\hat{P}(U_{\text{the}} = 1 c) \approx 1.0$

model), one for each term–class combination, rather than a number that is at least exponential in M , the size of the vocabulary. The independence assumptions reduce the number of parameters to be estimated by several orders of magnitude.

RANDOM VARIABLE C

To summarize, we generate a document in the multinomial model (Figure 13.4) by first picking a class $C = c$ with $P(c)$ where C is a random variable taking values from $\mathcal{C}$ as values. Next we generate term t_k in position k with $P(X_k = t_k|c)$ for each of the n_d positions of the document. The X_k all have the same distribution over terms for a given c . In the example in Figure 13.4, we show the generation of $\langle t_1, t_2, t_3, t_4, t_5 \rangle = \langle \text{Beijing, and, Taipei, join, WTO} \rangle$, corresponding to the one-sentence document *Beijing and Taipei join WTO*.

For a completely specified document generation model, we would also have to define a distribution $P(n_d|c)$ over lengths. Without it, the multinomial model is a token generation model rather than a document generation model.

We generate a document in the Bernoulli model (Figure 13.5) by first picking a class $C = c$ with $P(c)$ and then generating a binary indicator e_i for each term t_i of the vocabulary ($1 \leq i \leq M$). In the example in Figure 13.5, we show the generation of $\langle e_1, e_2, e_3, e_4, e_5, e_6 \rangle = \langle 0, 1, 0, 1, 1, 1 \rangle$, corresponding, again, to the one-sentence document *Beijing and Taipei join WTO* where we have assumed that and is a stop word.

We compare the two models in Table 13.3, including estimation equations and decision rules.

Naive Bayes is so called because the independence assumptions we have just made are indeed very naive for a model of natural language. The conditional independence assumption states that features are independent of each other given the class. This is hardly ever true for terms in documents. In many cases, the opposite is true. The pairs *hong* and *kong* or *london* and *en-*

► **Table 13.4** Correct estimation implies accurate prediction, but accurate prediction does not imply correct estimation.

	c_1	c_2	class selected
true probability $P(c d)$	0.6	0.4	c_1
$\hat{P}(c) \prod_{1 \leq k \leq n_d} \hat{P}(t_k c)$ (Equation (13.13))	0.00099	0.00001	
NB estimate $\hat{P}(c d)$	0.99	0.01	c_1

glish in Figure 13.7 are examples of highly dependent terms. In addition, the multinomial model makes an assumption of positional independence. The Bernoulli model ignores positions in documents altogether because it only cares about absence or presence. This bag-of-words model discards all information that is communicated by the order of words in natural language sentences. How can NB be a good text classifier when its model of natural language is so oversimplified?

The answer is that even though the *probability estimates* of NB are of low quality, its *classification decisions* are surprisingly good. Consider a document d with true probabilities $P(c_1|d) = 0.6$ and $P(c_2|d) = 0.4$ as shown in Table 13.4. Assume that d contains many terms that are positive indicators for c_1 and many terms that are negative indicators for c_2 . Thus, when using the multinomial model in Equation (13.13), $\hat{P}(c_1) \prod_{1 \leq k \leq n_d} \hat{P}(t_k|c_1)$ will be much larger than $\hat{P}(c_2) \prod_{1 \leq k \leq n_d} \hat{P}(t_k|c_2)$ (0.00099 vs. 0.00001 in the table). After division by 0.001 to get well-formed probabilities for $P(c|d)$, we end up with one estimate that is close to 1.0 and one that is close to 0.0. This is common: The winning class in NB classification usually has a much larger probability than the other classes and the estimates diverge very significantly from the true probabilities. But the classification decision is based on which class gets the highest score. It does not matter how accurate the estimates are. Despite the bad estimates, NB estimates a higher probability for c_1 and therefore assigns d to the correct class in Table 13.4. *Correct estimation implies accurate prediction, but accurate prediction does not imply correct estimation.* NB classifiers estimate badly, but often classify well.

CONCEPT DRIFT

Even if it is not the method with the highest accuracy for text, NB has many virtues that make it a strong contender for text classification. It excels if there are many equally important features that jointly contribute to the classification decision. It is also somewhat robust to noise features (as defined in the next section) and *concept drift* – the gradual change over time of the concept underlying a class like *US president* from Bill Clinton to George W. Bush (see Section 13.7). Classifiers like kNN (Section 14.3, page 297) can be carefully tuned to idiosyncratic properties of a particular time period. This will then hurt them when documents in the following time period have slightly

► **Table 13.5** A set of documents for which the NB independence assumptions are problematic.

- (1) He moved from London, Ontario, to London, England.
- (2) He moved from London, England, to London, Ontario.
- (3) He moved from England to London, Ontario.

different properties.

The Bernoulli model is particularly robust with respect to concept drift. We will see in Figure 13.8 that it can have decent performance when using fewer than a dozen terms. The most important indicators for a class are less likely to change. Thus, a model that only relies on these features is more likely to maintain a certain level of accuracy in concept drift.

NB's main strength is its efficiency: Training and classification can be accomplished with one pass over the data. Because it combines efficiency with good accuracy it is often used as a baseline in text classification research. It is often the method of choice if (i) squeezing out a few extra percentage points of accuracy is not worth the trouble in a text classification application, (ii) a very large amount of training data is available and there is more to be gained from training on a lot of data than using a better classifier on a smaller training set, or (iii) if its robustness to concept drift can be exploited.

In this book, we discuss NB as a classifier for text. The independence assumptions do not hold for text. However, it can be shown that NB is an *optimal classifier* (in the sense of minimal error rate on new data) for data where the independence assumptions do hold.

OPTIMAL CLASSIFIER

13.4.1 A variant of the multinomial model

An alternative formalization of the multinomial model represents each document d as an M -dimensional vector of counts $\langle \text{tf}_{t_1,d}, \dots, \text{tf}_{t_M,d} \rangle$ where $\text{tf}_{t_i,d}$ is the term frequency of t_i in d . $P(d|c)$ is then computed as follows (cf. Equation (12.8), page 243);

$$(13.15) \quad P(d|c) = P(\langle \text{tf}_{t_1,d}, \dots, \text{tf}_{t_M,d} \rangle | c) \propto \prod_{1 \leq i \leq M} P(X = t_i | c)^{\text{tf}_{t_i,d}}$$

Note that we have omitted the multinomial factor. See Equation (12.8) (page 243).

Equation (13.15) is equivalent to the sequence model in Equation (13.2) as $P(X = t_i | c)^{\text{tf}_{t_i,d}} = 1$ for terms that do not occur in d ($\text{tf}_{t_i,d} = 0$) and a term that occurs $\text{tf}_{t_i,d} \geq 1$ times will contribute $\text{tf}_{t_i,d}$ factors both in Equation (13.2) and in Equation (13.15).

```

SELECTFEATURES( $\mathbb{D}, c, k$ )
1   $V \leftarrow \text{EXTRACTVOCABULARY}(\mathbb{D})$ 
2   $L \leftarrow []$ 
3  for each  $t \in V$ 
4  do  $A(t, c) \leftarrow \text{COMPUTEFEATUREUTILITY}(\mathbb{D}, t, c)$ 
5      $\text{APPEND}(L, \langle A(t, c), t \rangle)$ 
6  return  $\text{FEATURESWITHLARGESTVALUES}(L, k)$ 

```

► **Figure 13.6** Basic feature selection algorithm for selecting the k best features.

?

Exercise 13.2

[★]

Which of the documents in Table 13.5 have identical and different bag of words representations for (i) the Bernoulli model (ii) the multinomial model? If there are differences, describe them.

Exercise 13.3

The rationale for the positional independence assumption is that there is no useful information in the fact that a term occurs in position k of a document. Find exceptions. Consider formulaic documents with a fixed document structure.

Exercise 13.4

Table 13.3 gives Bernoulli and multinomial estimates for the word *the*. Explain the difference.

13.5 Feature selection

FEATURE SELECTION

Feature selection is the process of selecting a subset of the terms occurring in the training set and using only this subset as features in text classification. Feature selection serves two main purposes. First, it makes training and applying a classifier more efficient by decreasing the size of the effective vocabulary. This is of particular importance for classifiers that, unlike NB, are expensive to train. Second, feature selection often increases classification accuracy by eliminating noise features. A *noise feature* is one that, when added to the document representation, increases the classification error on new data. Suppose a rare term, say *arachnocentric*, has no information about a class, say *China*, but all instances of *arachnocentric* happen to occur in *China* documents in our training set. Then the learning method might produce a classifier that misassigns test documents containing *arachnocentric* to *China*. Such an incorrect generalization from an accidental property of the training set is called *overfitting*.

NOISE FEATURE

OVERFITTING

We can view feature selection as a method for replacing a complex classifier (using all features) with a simpler one (using a subset of the features).

It may appear counterintuitive at first that a seemingly weaker classifier is advantageous in statistical text classification, but when discussing the bias-variance tradeoff in Section 14.6 (page 308), we will see that weaker models are often preferable when limited training data are available.

The basic feature selection algorithm is shown in Figure 13.6. For a given class c , we compute a utility measure $A(t, c)$ for each term of the vocabulary and select the k terms that have the highest values of $A(t, c)$. All other terms are discarded and not used in classification. We will introduce three different utility measures in this section: mutual information, $A(t, c) = I(U_t; C_c)$; the χ^2 test, $A(t, c) = X^2(t, c)$; and frequency, $A(t, c) = N(t, c)$.

Of the two NB models, the Bernoulli model is particularly sensitive to noise features. A Bernoulli NB classifier requires some form of feature selection or else its accuracy will be low.

This section mainly addresses feature selection for two-class classification tasks like *China* versus *not-China*. Section 13.5.5 briefly discusses optimizations for systems with more than two classes.

13.5.1 Mutual information

MUTUAL INFORMATION

A common feature selection method is to compute $A(t, c)$ as the expected *mutual information* (MI) of term t and class c .⁵ MI measures how much information the presence/absence of a term contributes to making the correct classification decision on c . Formally:

$$(13.16) \quad I(U; C) = \sum_{e_t \in \{1,0\}} \sum_{e_c \in \{1,0\}} P(U = e_t, C = e_c) \log_2 \frac{P(U = e_t, C = e_c)}{P(U = e_t)P(C = e_c)},$$

where U is a random variable that takes values $e_t = 1$ (the document contains term t) and $e_t = 0$ (the document does not contain t), as defined on page 266, and C is a random variable that takes values $e_c = 1$ (the document is in class c) and $e_c = 0$ (the document is not in class c). We write U_t and C_c if it is not clear from context which term t and class c we are referring to.

For MLEs of the probabilities, Equation (13.16) is equivalent to Equation (13.17):

$$(13.17) \quad I(U; C) = \frac{N_{11}}{N} \log_2 \frac{NN_{11}}{N_{1.}N_{.1}} + \frac{N_{01}}{N} \log_2 \frac{NN_{01}}{N_{0.}N_{.1}} \\ + \frac{N_{10}}{N} \log_2 \frac{NN_{10}}{N_{1.}N_{.0}} + \frac{N_{00}}{N} \log_2 \frac{NN_{00}}{N_{0.}N_{.0}}$$

where the N s are counts of documents that have the values of e_t and e_c that are indicated by the two subscripts. For example, N_{10} is the number of doc-

5. Take care not to confuse expected mutual information with *pointwise mutual information*, which is defined as $\log N_{11}/E_{11}$ where N_{11} and E_{11} are defined as in Equation (13.18). The two measures have different properties. See Section 13.7.

uments that contain t ($e_t = 1$) and are not in c ($e_c = 0$). $N_{1\cdot} = N_{10} + N_{11}$ is the number of documents that contain t ($e_t = 1$) and we count documents independent of class membership ($e_c \in \{0, 1\}$). $N = N_{00} + N_{01} + N_{10} + N_{11}$ is the total number of documents. An example of one of the MLE estimates that transform Equation (13.16) into Equation (13.17) is $P(U = 1, C = 1) = N_{11}/N$.



Example 13.3: Consider the class *poultry* and the term export in Reuters-RCV1. The counts of the number of documents with the four possible combinations of indicator values are as follows:

	$e_c = e_{\text{poultry}} = 1$	$e_c = e_{\text{poultry}} = 0$
$e_t = e_{\text{export}} = 1$	$N_{11} = 49$	$N_{10} = 27,652$
$e_t = e_{\text{export}} = 0$	$N_{01} = 141$	$N_{00} = 774,106$

After plugging these values into Equation (13.17) we get:

$$\begin{aligned}
 I(U;C) &= \frac{49}{801,948} \log_2 \frac{801,948 \cdot 49}{(49+27,652)(49+141)} \\
 &\quad + \frac{141}{801,948} \log_2 \frac{801,948 \cdot 141}{(141+774,106)(49+141)} \\
 &\quad + \frac{27,652}{801,948} \log_2 \frac{801,948 \cdot 27,652}{(49+27,652)(27,652+774,106)} \\
 &\quad + \frac{774,106}{801,948} \log_2 \frac{801,948 \cdot 774,106}{(141+774,106)(27,652+774,106)} \\
 &\approx 0.0001105
 \end{aligned}$$

To select k terms $t_1, \dots, t_k$ for a given class, we use the feature selection algorithm in Figure 13.6: We compute the utility measure as $A(t, c) = I(U_t, C_c)$ and select the k terms with the largest values.

Mutual information measures how much information – in the information-theoretic sense – a term contains about the class. If a term's distribution is the same in the class as it is in the collection as a whole, then $I(U; C) = 0$. MI reaches its maximum value if the term is a perfect indicator for class membership, that is, if the term is present in a document if and only if the document is in the class.

Figure 13.7 shows terms with high mutual information scores for the six classes in Figure 13.1.⁶ The selected terms (e.g., london, uk, british for the class *UK*) are of obvious utility for making classification decisions for their respective classes. At the bottom of the list for *UK* we find terms like peripherals and tonight (not shown in the figure) that are clearly not helpful in deciding

6. Feature scores were computed on the first 100,000 documents, except for *poultry*, a rare class, for which 800,000 documents were used. We have omitted numbers and other special words from the top ten lists.